

SYNOPSYS DESIGN COMPILER (DC) – RTL TO GATE-LEVEL

Project name : CHIP TOP

Synthesis is the process of converting RTL code (Verilog/VHDL) into a gate-level netlist using standard cells, while meeting timing, area, and power constraints.

• Synthesis Flow

This document explains a **complete Design Compiler synthesis script** step by step. It is written for

Learning and interview preparation, especially useful for VLSI freshers and early professionals.

• Environment Setup

```
cd vg_b4/chip_top/synth
```

```
csh
```

```
source /home/tools/synopsys/cshrc_synopsys
```

Explanation

- Moves to the synthesis working directory
- Switches to **csh shell** (required by Synopsys tools)
- Sources Synopsys environment variables and licenses

• Launch Design Compiler

```
dc_shell -output_log_file ./outputs/23_synth.log
```

Explanation

- Starts **dc_shell** (Design Compiler)
- Redirects tool messages to a log file for debugging and tracking

• Search Path Setup

```
set search_path "/home/rajshekhar/vg_b4/chip_top/synth/inputs/rtl"
```

Explanation

- Tells DC **where to find RTL files**
- Includes Verilog source code directories

• Target Library

```
set target_library "saed32hvt_ss0p95v125c.db saed32sraml_ss0p95v125c_iop95v.db  
saed32lvt_ss0p95v125c.db saed32sram_ss0p95v125c.db saed32rvt_ss0p95v125c.db"
```

Explanation

- Defines **standard cell and memory libraries**
- DC uses these cells to **optimize, map, and implement logic**
- Includes **HVT, LVT, RVT** cells for timing/power tradeoffs
- **Link Library**

```
set link_library "saed32hvt_ss0p95v125c.db saed32sraml_ss0p95v125c_iop95v.db
saed32lvt_ss0p95v125c.db saed32sram_ss0p95v125c.db saed32rvt_ss0p95v125c.db"
```

Explanation

- Used to **resolve references** during elaboration
- Must include **all libraries used by the design**
- Typically includes everything in target_library

- **Read RTL File List**

```
source ./inputs/rtl/rtl_file_list.tcl
```

Explanation

- Loads all RTL files using a centralized file list
- Helps maintain **clean and scalable projects**

- **Analyze RTL**

```
analyze -format verilog $top
```

Explanation

- Parses Verilog RTL
- Converts HDL into an **intermediate internal format**
- Syntax checking happens here

- **Elaborate Design**

```
elaborate ChipTop
```

Explanation

- Builds the **design hierarchy**
- Resolves module instances
- Prepares the design for synthesis

- **Apply Timing Constraints**

```
source ./inputs/rtl/my_chip.sdc
```

Explanation

- Applies **clock, IO delays, false paths, multicycle paths**
- Constraints guide optimization and QoR
- **Compile (High-Effort Optimization)**

```
compile_ultra -no_autoungroup
```

Explanation

- Performs **high-quality synthesis optimization**
- Improves timing, power, and area
- -no_autoungroup preserves hierarchy

- **Check Current Design**

```
current_design
```

Explanation

Confirms which design is active

Useful for sanity checks

- **Write Gate-Level Netlist**

```
write_file -format verilog -hierarchy -output ./outputs/chiptop.vg
```

Explanation

Generates **gate-level netlist**

-hierarchy keeps module structure intact

Output is used for P&R and STA

- **Generate Reports**

```
set rpt_dir ./outputs/qor_reports
```

```
# Create directory if it does not exist
```

```
if ![file exists $rpt_dir] {
```

```
    file mkdir $rpt_dir
```

```
}
```

```
report_power > $rpt_dir/power_report.rpt
```

```
report_timing > $rpt_dir/timing_report.rpt
```

```
report_area > $rpt_dir/area_report.rpt
```

Power Report → dynamic + leakage power

Global Operating Voltage = 0.95
Power-specific unit information :
Voltage Units = 1V
Capacitance Units = 1.000000ff
Time Units = 1ns
Dynamic Power Units = 1uW (derived from V,C,T units)
Leakage Power Units = 1pW

Attributes

i - Including register clock pin internal power

Cell Internal Power = -2.0730 mW (104%)
Net Switching Power = 83.5814 uW (-3%)

Total Dynamic Power = -1.9894 mW (100%)
Cell Leakage Power = 402.3449 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

Power Group	Internal Power	Switching Power	Leakage Power	Total Power (%) Attrs
io_pad	0.0000	0.0000	0.0000	0.0000 (-0.00%)
memory	-2.5304e+03	3.0625	0.0000	-2.5273e+03 (159.24%)
black box	0.0000	0.0000	0.0000	0.0000 (-0.00%)
clock network	0.0000	0.0000	0.0000	0.0000 (-0.00%) i
register	0.0000	0.0000	0.0000	0.0000 (-0.00%)
sequential	341.6599	0.6634	1.7059e+08	512.9169 (-32.32%)
combinational	115.7089	79.8555	2.3175e+08	427.3154 (-26.92%)
Total	-2.0730e+03 uW	83.5814 uW	4.0234e+08 pW	-1.5871e+03 uW

1

Timing Report

Report : timing
-path full
-delay max
-max paths 1
Design : ChipTop
Version: V-2023.12-SP5
Date : Sat Dec 27 13:23:36 2025

Operating Conditions: ss0p95v125c Library: saed32hvt_ss0p95v125c
Wire Load Model Mode: enclosed

Startpoint: InstDecode/WE_X_Dly1_reg
(rising edge-triggered flip-flop)
Endpoint: MemWriteBus[63]
(output port)
Path Group: (none)
Path Type: max

Des/Clust/Port	Wire Load Model	Library
InstructionDecoder	8000	saed32hvt_ss0p95v125c
ChipTop	540000	saed32hvt_ss0p95v125c
address_gen_add	8000	saed32hvt_ss0p95v125c
MemGHier	280000	saed32hvt_ss0p95v125c

Point	Incr	Path
InstDecode/WE_X_Dly1_reg/CLK (DFFSSRX1 HVT)	0.00	0.00 r
InstDecode/WE_X_Dly1_reg/Q (DFFSSRX1 HVT)	0.30	0.30 f
InstDecode/WrtEnbX (InstructionDecoder)	0.00	0.30 f
AddressGen/write X enable (address_gen_add)	0.00	0.30 f
AddressGen/U45/Y (OR2X1 HVT)	0.14	0.43 f
AddressGen/U112/Y (INVX0 HVT)	0.09	0.52 r
AddressGen/U120/Y (AO22X1 HVT)	0.13	0.65 r
AddressGen/common_address[7] (address_gen_add)	0.00	0.65 r
MemGHier/common_address[7] (MemGHier)	0.00	0.65 r
MemGHier/MemXb/0[0] (SRAML1RW64x32)	0.05	0.69 r
MemGHier/U1/Y (IBUFFX4 LVT)	0.05	0.74 f
MemGHier/U2/Y (INVX1 LVT)	0.02	0.76 r
MemGHier/memX_dataout[0] (MemGHier)	0.00	0.76 r
AddressGen/memX_datain[0] (address_gen_add)	0.00	0.76 r
AddressGen/U25/Y (INVX1 RVT)	0.02	0.78 f
AddressGen/U24/Y (INVX0 HVT)	0.02	0.80 r
AddressGen/U46/Y (NAND2X0 HVT)	0.07	0.87 f
AddressGen/U48/Y (INVX1 HVT)	0.05	0.91 r
AddressGen/U111/CO (FADDX1 HVT)	0.13	1.05 r
AddressGen/U110/CO (FADDX1 HVT)	0.14	1.19 r
AddressGen/U109/CO (FADDX1 HVT)	0.14	1.33 r
AddressGen/U108/CO (FADDX1 HVT)	0.14	1.47 r
AddressGen/U107/CO (FADDX1 HVT)	0.14	1.62 r
AddressGen/U106/CO (FADDX1 HVT)	0.14	1.76 r
AddressGen/U105/CO (FADDX1 HVT)	0.14	1.90 r
AddressGen/U104/CO (FADDX1 HVT)	0.14	2.04 r
AddressGen/U103/CO (FADDX1 HVT)	0.14	2.18 r
AddressGen/U102/CO (FADDX1 HVT)	0.14	2.33 r
AddressGen/U101/CO (FADDX1 HVT)	0.14	2.47 r
AddressGen/U100/CO (FADDX1 HVT)	0.14	2.61 r
AddressGen/U99/CO (FADDX1 HVT)	0.14	2.75 r
AddressGen/U98/CO (FADDX1 HVT)	0.14	2.89 r
AddressGen/U97/CO (FADDX1 HVT)	0.14	3.04 r
AddressGen/U96/CO (FADDX1 HVT)	0.14	3.18 r
AddressGen/U95/CO (FADDX1 HVT)	0.14	3.32 r
AddressGen/U94/CO (FADDX1 HVT)	0.14	3.46 r
AddressGen/U93/CO (FADDX1 HVT)	0.14	3.60 r
AddressGen/U92/CO (FADDX1 HVT)	0.14	3.75 r

Area Report → standard cell utilization

```
*****
Report : area
Design : ChipTop
Version: V-2023.12-SP5
Date   : Sat Dec 27 12:21:56 2025
*****
```

Library(s) Used:

```
saed32hvt_ss0p95v125c (File: /home/rajshekhars/vg_b4/chip_top/synth/inputs/rtl/saed32hvt_ss0p95v125c.db)
saed32lvt_ss0p95v125c (File: /home/rajshekhars/vg_b4/chip_top/synth/inputs/rtl/saed32lvt_ss0p95v125c.db)
saed32sramlp_ss0p95v125c_i0p95v (File: /home/rajshekhars/vg_b4/chip_top/synth/inputs/rtl/saed32sramlp_ss0p95v125c_i0p95v.db)
saed32rvt_ss0p95v125c (File: /home/rajshekhars/vg_b4/chip_top/synth/inputs/rtl/saed32rvt_ss0p95v125c.db)
```

```
Number of ports:      3814
Number of nets:       8084
Number of cells:      4743
Number of combinational cells: 3953
Number of sequential cells:  710
Number of macros/black boxes:  4
Number of buf/inv:     973
Number of references:   10
```

```
Combinational area:      10417.616684
Buf/Inv area:           1632.621071
Noncombinational area:   5190.636991
Macro/Black Box area:    47770.062500
Net Interconnect area:   5986.556165
```

```
Total cell area:        493378.316174
Total area:              499364.872339
1
```

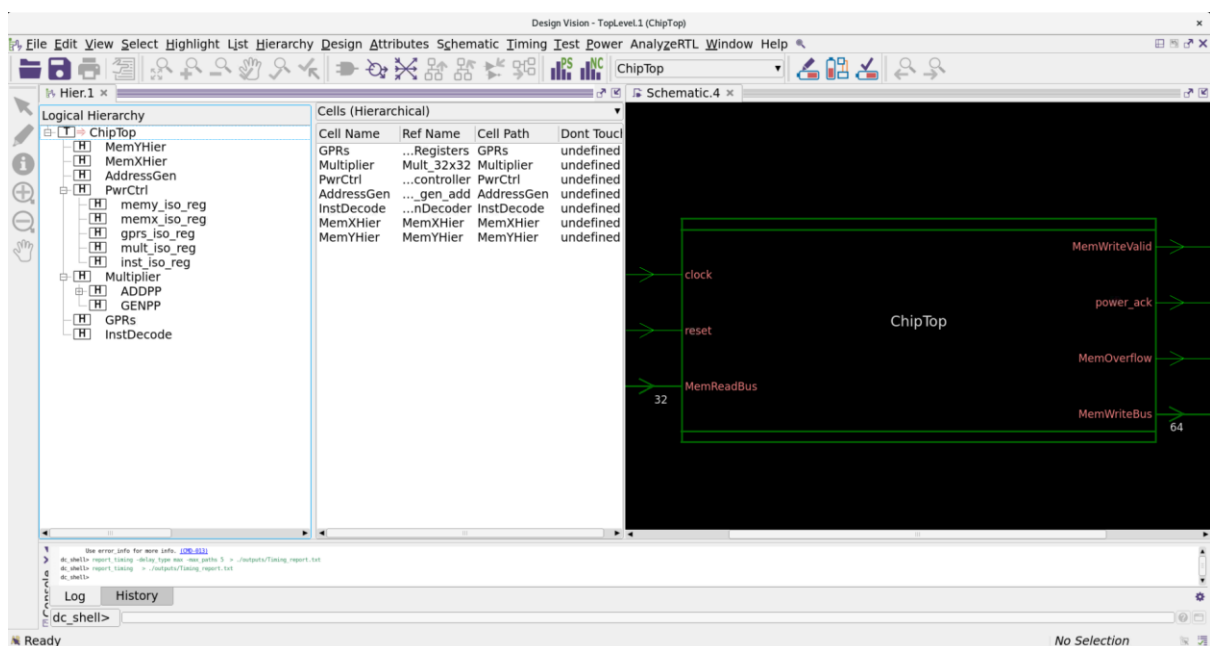
- **Launch GUI (Optional)**

start_gui

Explanation

Opens Design Compiler GUI

Useful for schematic, timing, and constraint visualization



Design Vision - TopLevel1 (ChipTop)

File Edit View Select Highlight List Hierarchy Design Attributes Schematic Timing Test Power AnalyzeRTL Window Help

ChipTop

Hier.1 x Schematic.4 x

Log History

dc_shell>

Click objects or drag a box to select (Hold Ctrl to add, Shift to remove) Alt DoubleClick expands Fanin/Out ALL levels

Design: ChipTop

RealVNC Ser... rightclick... placement.t... final_placem... checks_affe... voltage.tcl +... script.tcl + (... power_plan... power_plan... flow.tcl + (... Design Visto... Man Page VL... chip_top.fl... Timing_repo... 1/2